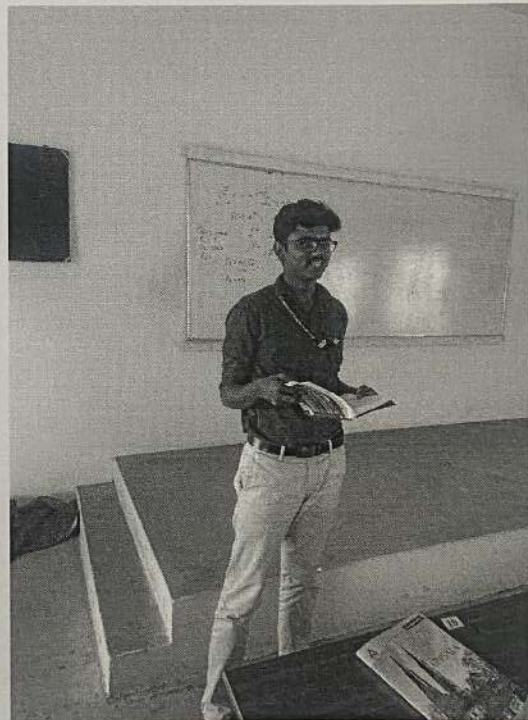
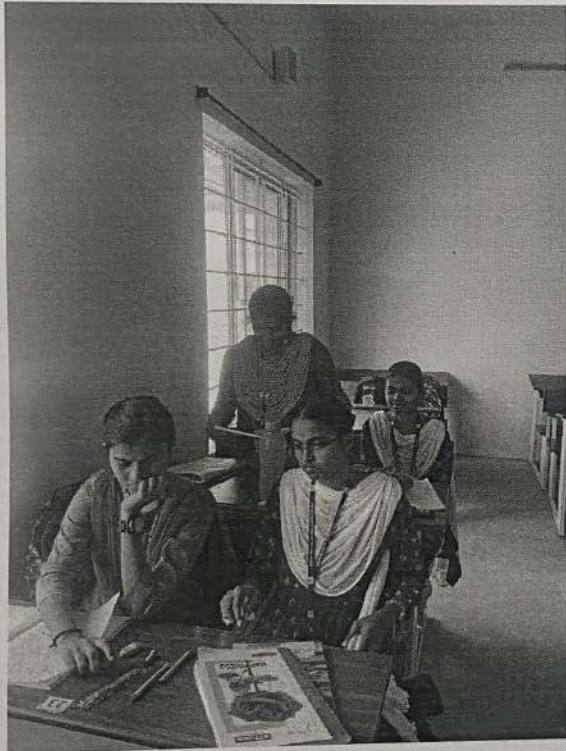
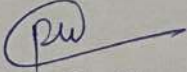


RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

Degree, Semester & Branch : III Semester B.E. ECE A
Course Code & Title : EC3351 Control Systems
Name of the Faculty member : P.Venkatesh
Name of the topic : Basic structure, feedback theory
Name of the event : Class poll




Signature of the Faculty member


HOD

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

Degree, Semester & Branch : III Semester B.E. ECE A

Course Code & Title : EC3351 Control Systems

Name of the Faculty member : P.Venkatesh

Name of the topic : Steady error constant – static error & dynamic error

Name of the event : Quiz

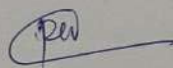
Quiz M. Karthika

Various Types of Inputs

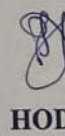
Input signal	Type number of system			
	0	1	2	3
Unit step	$\frac{1}{1+K_p}$	0	0	0
Unit Ramp	∞	$\frac{1}{K_v}$	0	0
Unit Parabolic	∞	∞	$\frac{1}{K_a}$	0

Unit-2

- 1) $V_p = \frac{H}{s^2} \cos(s\omega) H(s)$
- 2) $V_p = \frac{H}{s^2} \sin(s\omega) H(s)$
- 3) $E_s(t) = \frac{1}{H+P}$ for unit step
- 4) Input is not defined
- 5) Taylor series



Signature of the Faculty member


HOD



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III semester B.E. ECE A

Course Code & Title: EC8391 & Control Systems Engineering

Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** IV / Problems in root locus
- **Course Outcome:** CO4 : The students will be able to determine the stability of control systems using various methods
- **Topic Learning Outcome:** TLO 14: To determine the stability of the given system using Nyquist Stability Criterion
- **Activity Chosen:** Jigsaw

- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO1
CO4	3	3	2	1	1	1	2

- **PO / PSO mapped:**

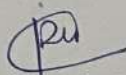
Innovative practice	PO1	PO2	PO3	PO8	PO9	PO10	PSO1
	3	3	2	1	1	1	2
Justification for correlation	Determining the stability of a given system requires strong knowledge in advanced mathematical	The students will analyze the complex engineering problems and arrive	This PO is mapped to level 3 because students will refine	During preparation of assignment and report/feedback on collaborative	The students will function effectively as a team to be	Presentation of the results and conclusions of the activity carried out, by the students,	This PSO has been mapped with level 2 as the methods

❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ https://www.educationworld.com/a_curr/strategy/strategy036.shtml
- <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>



Signature of Faculty Member



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